

- A tamper detection can optionally erase the backup registers, SRAM2, ICACHE, and cryptographic peripherals. The device resources protected by tamper are named "device secrets".
- 32 × 32-bit backup registers
- Up to three tamper pins for three external tamper detection events:
 - Passive tampers: Ultralow-power edge or level detection with internal pull-up hardware management
 - Configurable digital filter
- Six internal tamper events to protect against transient attacks
- Each tamper can be configured in two modes:
 - Confirmed mode: immediate erase of secrets on tamper detection, including backup registers erase
 - Potential mode: most of the secrets erase following a tamper detection are launched by software
- Any tamper detection can generate an RTC timestamp event
- Tamper configuration and backup registers privilege protection

3.22 Inter-integrated circuit interface (I2C)

The device embeds two I2C interfaces. Refer to [Table 7](#) for feature implementation.

The I²C bus interface handles communications between the microcontroller and the serial I²C bus. It controls all I²C bus-specific sequencing, protocol, arbitration, and timing.

It supports Standard-mode (Sm), Fast-mode (Fm), and Fast-mode Plus (Fm+).

The I2C peripheral is also system management bus (SMBus) and power management bus (PMBus[®]) compatible. It can use DMA to reduce the CPU load.

The I²C peripheral supports:

- I²C-bus specification rev03 compatibility:
 - Controller and target modes
 - Multicontroller capability
 - Standard-mode (up to 100 kHz)
 - Fast-mode (up to 400 kHz)
 - Fast-mode Plus (up to 1 MHz)
 - 7-bit and 10-bit addressing mode
 - Multiple 7-bit target addresses (2 addresses, 1 with configurable mask)
 - All 7-bit addresses acknowledge mode
 - General call
 - Programmable setup and hold times
 - Easy-to-use event management
 - Clock stretching (optional)
- 1-byte buffer with DMA capability
- Programmable analog and digital noise filters
- SMBus specification rev 3.0 compatibility:
 - Hardware PEC (packet error checking) generation and verification with ACK control
 - Command and data acknowledge control
 - Address resolution protocol (ARP) support
 - Host and device support – SMBus alert
 - Timeouts and idle condition detection
- PMBus rev 1.3 standard compatibility
- Independent clock
- Wake-up from Stop mode on address match

Table 7. I2C implementation

Features ⁽¹⁾	I2C1	I2C2
Standard-mode (up to 100 Kbit/s)	X	X